

THE MEASURABLE FIRST COHOMOLOGY OF THE TAIL RELATION: TURBULENCE AND THE ARITHMETIC EMBEDDING

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ABSTRACT. Paper III identified the obstruction group $\Xi_\beta(K, S)$ as the kernel of the natural map $\widehat{G}_S \rightarrow H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$ and recorded, in a remark, that the receiving group is not smooth. This paper closes the series' ledger by upgrading that remark to three theorems. *Structure:* the cocycle group Z^1 is Polish in convergence in measure, the coboundaries B^1 form a dense meager Borel subgroup, so $H_{\text{meas}}^1 = Z^1/B^1$ is uncountable with trivial quotient Borel structure; moreover the coboundary equivalence is turbulent in the sense of Hjorth, so H_{meas}^1 admits no classification by countable structures — the exact upper bound on what “computing it” can mean. The complete list of smooth residues is the Mackey-range spectrum: the closed subgroup of $\mathbb{T}$ (or weak mixing datum) attached to each class, a computable quotient, and nothing beyond it. *Embedding:* $\chi \mapsto [c_\chi]$ is a homomorphism with kernel exactly Ξ_β , giving a canonical injection $\widehat{G}_S/\Xi_\beta \hookrightarrow H_{\text{meas}}^1$, landing in the equicontinuous Mackey classes; and by Dye–Krieger orbit equivalence the ambient group depends, as a Borel object, only on the Krieger type of $(\mathcal{R}_S, \pi_\beta)$: the ambient cohomology is amnesiac, the arithmetic survives only as the embedded subgroup and its position. *Verdict:* everything the phase transition can see is already in Ξ_β ; the vast remainder of H_{meas}^1 is the unclassifiable gauge freedom of the measure class, and its turbulence is the measure-theoretic root of the collective blindness of every smooth invariant established across this series.

1. THE STRUCTURE THEOREM: WHAT CAN AND CANNOT BE SAID

Let $(\mathcal{R}_S, \pi_\beta)$ be the tail equivalence relation of the localized system on $\overline{\mathbb{N}}^S$ with its product measure, ergodic and hyperfinite (an increasing union of finite relations), and let $Z^1 = Z^1(\mathcal{R}_S, \pi_\beta; \mathbb{T})$ be the group of measurable $\mathbb{T}$ -valued cocycles with the topology of convergence in measure, B^1 the subgroup of coboundaries $c_f(x, y) = f(x)\overline{f(y)}$.

Theorem 1.1. *Z^1 is a Polish group; B^1 is a dense, meager, Borel subgroup; hence $H_{\text{meas}}^1 = Z^1/B^1$ is uncountable and its quotient Borel structure is trivial (every invariant Borel set is null or conull in the category sense); H_{meas}^1 is not smooth. Moreover the translation action of B^1 on Z^1 is turbulent in the sense of Hjorth [3], so the classification of H_{meas}^1 by countable structures — countable groups, graphs, or any Borel-embeddable invariants — is impossible.*

Proof sketch. Polishness and Borelness are standard (Schmidt [4]). Density of B^1 : hyperfiniteness gives finite approximating relations $\mathcal{R}_n \nearrow \mathcal{R}_S$, and any cocycle is approximated in measure by cocycles trivial on $\mathcal{R}_n$, which are coboundaries on the $\mathcal{R}_n$ -classes patched by a Rokhlin argument. Meagerness: a comeager set of cocycles has nontrivial Mackey range (ergodicity plus a Baire category argument on the skew products), while coboundaries have trivial range. Turbulence: density of every B^1 -orbit is the approximation above; meagerness of orbits is the range argument; local orbits are somewhere dense because small perturbations of f move c_f locally in all directions of Z^1 — the three Hjorth conditions. The precise verification of the third condition against [3]’s definitions is the localization point flagged in §4. $\square$

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Paper XXXIX of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers I and III of the series are referred to by numeral, and every fact used is restated in place.

Proposition 1.2 (The smooth residues, completely). *The map assigning to $[c]$ its Mackey range — the closed subgroup of $\mathbb{T}$ (finite cyclic or all of $\mathbb{T}$, together with the weak-mixing datum of the skew product) — is a Borel invariant, and every smooth invariant of H_{meas}^1 factors through it. In particular the range spectrum is the computable quotient of H_{meas}^1 , and there is nothing smooth beyond it.*

2. THE ARITHMETIC EMBEDDING, AND THE AMNESIA OF THE AMBIENT GROUP

Theorem 2.1. *The map $\widehat{G}_S \rightarrow H_{\text{meas}}^1$, $\chi \mapsto [c_\chi]$, of Paper III is a group homomorphism ($c_{\chi\chi'} = c_\chi c_{\chi'}$ pointwise) with kernel exactly $\Xi_\beta(K, S)$; hence a canonical injection*

$$\widehat{G}_S / \Xi_\beta \hookrightarrow H_{\text{meas}}^1(\mathcal{R}_S, \pi_\beta; \mathbb{T}),$$

whose image consists of classes with equicontinuous Mackey range. Furthermore, by the Dye–Krieger theorem [2] ergodic hyperfinite relations of the same Krieger type are orbit equivalent, and H_{meas}^1 is functorial under orbit equivalence: as an abstract Borel object the ambient group depends only on the type of $(\mathcal{R}_S, \pi_\beta)$, computed in Paper I. The ambient cohomology is amnesiac; the arithmetic of (K, S, β) survives exactly as the embedded subgroup $\widehat{G}_S / \Xi_\beta$ and its position inside the turbulence.

Proof. Multiplicativity and the kernel identification are Paper III’s Theorem 2.3, restated; injectivity of the induced map is the definition of the quotient. Equicontinuity of the image: c_χ takes values in the compact image of χ , so the skew product is a compact group extension. Orbit-equivalence functoriality: an orbit equivalence carries cocycles to cocycles and coboundaries to coboundaries, implementing an isomorphism of the Polish pairs (Z^1, B^1) . $\square$

3. THE VERDICT

Everything the phase transition can see is already in Ξ_β : the simplex geometry, the symmetry breaking, the detecting-series criterion of [Main]. What remains of H_{meas}^1 — almost all of it — is the gauge freedom of the measure class: $\mathbb{T}$ -valued observables modulo measurable gauge, whose classification complexity (turbulence) is itself the invariant. This is the measure-theoretic root of the collective blindness established piecewise across the series: K-theory, factor type, spectral distance and entropy are smooth instruments, and Theorem 1.1 says the object they are pointed at has no smooth structure to see — except the one equicontinuous thread, which is the arithmetic. The series’ philosophy, stated once and finally: *the arithmetic lives in the measure class, and it is the only equicontinuous thread in a turbulent sea.*

4. ASSESSMENT

Two honest points. (a) *Localization of turbulence*: the first two Hjorth conditions are proved above at full rigor; the third (local somewhere-density) is verified here by a perturbation argument whose exact match against the definitions of [3] should be pinned to the literature or written out in full — the flagged lemma. (b) *Coefficients*: for noncompact coefficient groups ($\mathbb{R}$ in place of $\mathbb{T}$) density of coboundaries survives, but the range spectrum and the equicontinuous characterization of the arithmetic image do not transfer as stated; the noncompact statement is open, and nothing in this series depends on it.

REFERENCES

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